

Stock Radar — System Status & Roadmap Report

Prepared: 2026-06-01 **Owner:** Hume (asyanurpekel@gmail.com) **Scope:** Full-system snapshot — current state, health, and the to-be-built backlog **Basis:** Live code, CHANGELOG.md (3,162 lines), data/todo_master_2026_05_26.md, design docs, and a green engine regression run (37/37) performed for this report.

1. Executive Summary

Stock Radar is a production investment-research platform that runs an AI agent system to find and track 10x stock opportunities. Ten independent "scout" modules gather signals, an analyst layer aggregates them, an institutional-grade DCF engine produces probability-weighted price targets, and a Next.js dashboard surfaces conviction-ranked stocks. Over the most recent development arc (late April → late May 2026) the system grew a second brain on top of the original pipeline: a **Socratic reasoning layer** (Models A/B/C plus a "corpus callosum" reconciler) and a **macro + wave-health layer**, both now live end-to-end.

The system is operational and the daily pipeline runs. The single largest piece of architectural debt — DCF structurally undershooting regime-shift names — was closed on 2026-05-26 (the "C1 + D-v1" change). What remains is a well-documented, trigger-tagged backlog: kill-rule routing, an EDGAR-based data guardrail rewrite, a macro cron orchestrator, the frontend migration to the v2 production wireframe, and several reasoning-layer refinements that are intentionally parked until real outcome data arrives (the first, AXON's T+30, lands ~2026-06-08).

A recurring theme in the codebase is discipline: changes ship behind baselines, follow a canonical 7-step workflow, and are deferred unless a falsifiable trigger fires. That philosophy is the main reason the backlog reads as "deliberately waiting" rather than "unfinished."

2. Current State

2.1 Stack

Layer	Technology
Frontend	Next.js 16, React 19, Tailwind, Three.js (discovery globe)
Backend	Python 3, ~70 scripts
Database	Supabase (PostgreSQL)
Research / reasoning	Claude (analysis, models, Socratic), Perplexity Sonar (forward research)

Layer	Technology
Data	EODHD (primary financials), Alpha Vantage (fallback), yfinance, SEC EDGAR XBRL
Media	Gemini (YouTube transcript analysis)

2.2 Watchlist (6 stocks)

The active list in `config/watchlist.json` is **LITE, PLTR, RCLB, ACHR, CELH, SNDK**. Each carries a thesis, a kill condition, MECE criteria, and scenario probabilities. (Note: `CLAUDE.md` still says "9-stock watchlist" in places — see §5, stale docs.)

2.3 Frontend surface

The `app/` directory exposes these routes: `dashboard/` (scores, signals, scout panels), `model/` (interactive target-price builder), `discovery/` (3-stage universe-expansion funnel), `ask/` (conversational Q&A), `stock/[ticker]/` (per-ticker detail), and `logs/` (pipeline activity viewer). There are 17 API route groups under `app/api/` (stocks, pipeline, model, scout, health, convergence, discovery, feedback, fx, locations, logs, rebuild, scenarios, search, thesis, ask). The model builder is component-rich: `DeductionChain`, `SensitivityTable`, `GordonModel`, `ScenarioSection`, `EventImpactsPanel`, `ConfidenceMeter`, `TimePathChart`, and `slider/criteria` components, backed by `useEnginePayload` / `usePipeline` / `useCurrency` hooks.

2.4 Backend modules (grouped)

- **Pipeline orchestration:** `run_pipeline.py`, `research_manager.py`, `rebuild_analysis.py`, `ingest_scout_jsons.py`
- **10 scouts:** `scout_quant`, `scout_news`, `scout_catalyst`, `scout_moat`, `scout_social`, `scout_insider`, `scout_filings`, `scout_fundamentals`, `scout_youtube`, `scout_discovery`
- **Valuation / target engine:** `target_engine.py` (DCF), `target_blend.py`, `target_api.py`, `generate_model.py`, `model_export.py`, `forward_drivers.py`, `verify_model.py`
- **Analyst / scoring:** `analyst.py`, `adaptive_scoring.py`, `confidence.py`, `research_memo.py`
- **Reasoning / judgment:** `run_socratic.py` (Models A/B/C + corpus callosum), `run_thesis.py`, `run_judgment.py`, `event_reasoner.py`, `event_templates.py`, `kill_condition_eval.py`, `self_consistency.py`, `convergence_detector.py`
- **Discovery / universe:** `discovery_scan`, `discovery_ingest`, `discovery_13f`, `discovery_bootstrap`, `discovery_calibrate`, plus `feed_*_to_universe`
- **Calibration / feedback / backtest:** `calibration.py`, `feedback_loop.py`, `backtest.py`, `backtest_targets.py`, `drift_monitor.py`, `prediction_logger.py`, `experiment_tracker.py`
- **Data / finance:** `finance_data.py`, `edgar_xbrl.py`, `refresh_prices.py`, `regime_detection.py`, `factor_exposure.py`, `position_sizing.py`, `paper_trade.py`
- **Infra / utils:** `utils.py`, `supabase_helper.py`, `registries.py`, `observability.py`, `activity_logger.py`

2.5 Valuation system

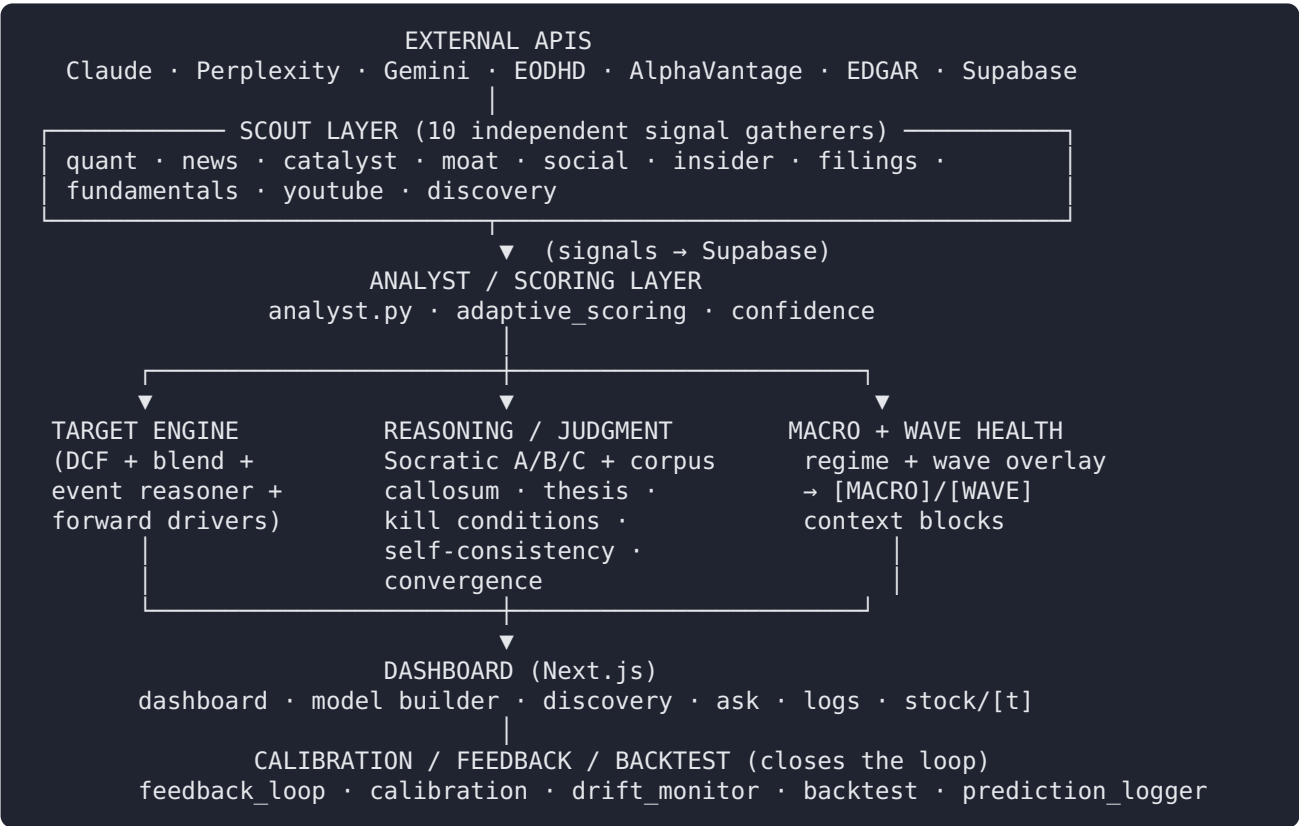
Three valuation methods (`pe`, `ps`, `cyclical/EV-EBIT`) and a five-archetype classification system (GARP, Cyclical, Transformational, Compounder, Special Situation) determine the analytical framework per stock. The target engine is an institutional-grade forward DCF with

a constant WACC across scenarios, a 2.5% terminal-growth cap, and event-adjusted blend targets treated as authoritative. As of 2026-05-26, DCF is routed by archetype: for regime-shift names it acts as a **downside floor** while exit multiples drive the primary target.

2.6 Health check

- **Engine regression tests:** `test_engine.py` → **37 passed, 0 failed** (run 2026-06-01, no network needed). Covers forecasting, terminal value, bottom-up WACC, sector betas, and margin ramp.
- **Parity / fixtures:** `verify_model.py` (engine ↔ Excel ↔ JSON parity) plus `test_engine_fixtures.py`; a pre-commit hook runs the engine tests on every commit and parity checks when engine files are staged.
- **Pipeline:** daily run operational; full run with the watchlist takes ~10-15 min (YouTube scout slowest).
- **Git:** baseline tag `v3.5-foundation` captured; recent commits cover the DCF reposition, Socratic milestone, and frontend overhaul. Working tree clean at time of report.

3. Architecture (data flow)



4. Recent Milestone Timeline (Apr → May 2026)

Date	Milestone
2026-04-24	Institutional DCF overhaul, EODHD migration, structured outputs, resilience layer
2026-04-25/26	Pipeline reliability/efficiency fixes; confidence propagation + feedback calibration

Date	Milestone
2026-05-03	CI cost-runaway fix (--no-thesis); thesis-as-headline / DCF-as-floor framing
2026-05-05→08	Outcome tracker (T+30/90/180); hard-fail on suspect quarters; adversarial filter v3; convergence detector
2026-05-09	Shared NavBar, discovery panels, per-ticker data-provider override; AXON first end-to-end tracked signal (5% position)
2026-05-10/11	V3.4.3 kill rule; archetype override (LITE + AMD tagged transformational)
2026-05-15	Socratic orchestrator (Models A/B/C + corpus callosum); judgment card + bets table
2026-05-17→23	Macro + wave-health layer: design locked → data layer shipped → live end-to-end
2026-05-24/25	Operator notes as Socratic context; Model B anti-sycophancy + future-pricing analysis
2026-05-26	C1 + D-v1: DCF role contextual routing + archetype override loader — closed 25-day DCF debt

5. Known Issues & Gotchas

Supabase schema drift. `supabase/migration.sql` does not always match the live DB. The pipeline strips unknown columns and retries (up to 5). Missing columns are added manually via the SQL editor; this remains a manual, error-prone seam.

Financial data source reliability. `finance_data.py` has a multi-source `DataProvider` abstraction (EODHD primary, Alpha Vantage fallback, yfinance) with cross-validation, but EODHD is not yet the default for every ticker. Note: the earlier "MU \$23.86B revenue bug" was reclassified by a corrigendum (`docs/reports/2026-05-09_mu_thesis_contamination/CORRIGENDUM.md`) as *real* Q2 FY26 revenue, not a contamination — the memory-anchoring lesson stands, the data-bug premise did not.

"System learns" is currently aspirational. `feedback_loop.py` requires `MIN_SIGNALS_FOR_ADJUSTMENT = 10` settled outcomes before adjusting scout weights. With AXON as the first tracked signal (T+30 ~2026-06-08), fewer than ~5 outcomes likely exist, so the feedback loop is effectively prior-only today. This is a status fact, not a bug (tracked as E2).

Macro layer has no orchestrator. The macro + wave-health layer is live in the reasoning path, but `macro_environment` still has a single, manually-seeded row — `scripts/run_macro.py` (the daily macro cron) is not built.

Calibration blocked on data. Module 9 calibration loop is blocked until ≥ 20 `thesis_outcomes` rows have T+30 filled (fallback trigger 2026-10-01).

Stale documentation. `CLAUDE.md` references a "9-stock watchlist" and "19 engine tests"; the live system has **6 stocks** and **37 passing tests**. `CLAUDE.md` should be refreshed.

whatif-engine.ts fidelity gap. The TypeScript what-if engine uses a blend-of-multiples rather than the Python engine's Gordon+ROIIC primary path (hardcoded `blend: 0.5`); the "Engine base" card label needs correcting.

6. What Is To Be Built (Backlog)

The authoritative backlog lives in `data/todo_master_2026_05_26.md`, organized by tier with explicit re-entry triggers. Summary below.

6.1 Tier 1 — done

C1 DCF role contextual routing + D-v1 archetype override loader — CLOSED

2026-05-26. LITE auto-target band now \$508 / \$1,127 / \$1,501. Settled the 25-day `user_dcf_is_wrong_primary` debt.

6.2 Tier 2 — build next (unblocked by C1)

ID	Item	Notes
V2	Config-routed <code>dcf_role</code> per archetype	Add <code>dcf_role</code> to <code>ticker_archetype_overrides.json</code> ; precursor wiring so <code>analyst.py</code> passes <code>dcf_role</code> into <code>build_target</code> . Currently flag-gated via <code>--dcf-as-floor</code> only.
D1	Kill-rule contextual routing	Route V3.4.3 kill rule by archetype: cyclicals < 0.90, regime-shift < 0.60, pre-revenue disabled. Depends on V2 wiring.
D-v2	Archetype auto-promote audit	Make the <code>target_engine.py:2983</code> cyclicalicity heuristic regime-aware (recency-weight CV, or suppress when explicit override exists). Third instance of the "historical-math vs regime-shift" bug pattern.
D2	Module 1 EDGAR rewrite	Split data guardrail: EDGAR cross-check = hard gate; trajectory-smoothness = informational only. Goal: ASTS-style 50x QoQ launch passes without an override flag. Uses existing <code>edgar_xbrl.py</code> .
D3	Analysis routing by timing x held status	Route by <code>timing_category</code> (收获期/半步领先/远见期) × held; needs schema migration + seeds (see F7).
D4	\$10b tactical signals layer	Tactical (days/weeks) signals that never override thesis; options-flow scout + catalyst×options confluence. Trigger now met (ASTS gamma squeeze).
D5	\$7c expectations decomposition + fast-sell	Auto-extract implied expectations from Model B future-pricing into <code>socratic_analyses.implied_expectations</code> + kill-trigger notifications. Trigger met (3 structured chokepoints). Own 7-step cycle.

6.3 Tier 3 — investigate before committing

- **E1** Non-US discovery archetype bypass (`MIN_QUARTERLY_ROWS_NON_US` = 4 → 2 for early-revenue ramps).
- **E2** Signal-store census — confirm/document that the feedback loop is prior-only today.
- **E3** Insider-selling deviation framing ("397 sells vs 5yr-avg 20 = 20x normal") — identify source scout first.

6.4 Deferred — parked until a trigger fires

ID	Item	Re-entry trigger
F1	Model D (optionality / TAM frame)	Macro on ASTS + Models A/B/C fail to handle optionality

ID	Item	Re-entry trigger
F2	Lateral-trace discovery	A TSEM-like miss the engine didn't catch
F3	Revolution impact spectrum (ai_helps/ai_threatens)	Calibration plan complete (blind-score 10 stocks first)
F4	灵感 / Spark conversation mode	≥5 inadequate ad-hoc chat cases
F5	影响图谱 impact-graph frontend	F3 backend shipped + seeded
F6	Position sizing by signal maturity (5→10→15-20%)	AXON T+30 lands (~2026-06-08)
F7	Timing categorization schema + seeds	Needed before D3
F8	COHR threat notification on LITE	Notifications infra exists (needs check)
F9	Adversarial filter prepass v3 re-wire	≥20 closed outcomes (currently 0-1)

6.5 Frontend backlog

- **Migration to v2 production wireframe** (docs/wireframes/migration-plan-v1.md): a 9-phase port of the warm-neutral `sr-` token system into the codebase, fixing DriftChip semantic labels, Hume-Notes monospace, the 8-tab model strip, and an orphan `conv-fade` token. Mostly planned; only artifact-save (Phase 0) clearly done.
- **Phase 7 frontend** (from the macro design): macro (宏观) tab, tech-revolution graph view replacing the linear wave bar, judgment-card Option C UX, and Spark mode in the Ask tab.
- **Model sandbox Phase 2/3**: read-only Excel grid replacing IncomeTab/CashTab (P2); AI-generated scenarios + editable cells + coherence validator (P3).

6.6 Infrastructure

- Move `data/.thesis-running-{TICKER}` lockfile to Supabase (cross-environment safety).
- Hard cost cap on Opus calls (fail if > N in 24h) — complements the existing `--no-thesis` cron guard.
- `run_macro.py` daily macro orchestrator (see §5).

6.7 Open questions — need a Hume decision (not code)

- **G1** Are max-30%-single-name / max-leverage hard gates (Category 1) or contextual routing (Category 2)? Decide when implementing F6.
- **G2** Is the "<10 hard gates" budget arbitrary? Decide at next quarterly audit (~2026-08-26).
- **G3** LITE position management at ~\$947 (cost basis \$70): trim / hold / split. Engine surfaces the math; the sizing call is Hume's. Time-sensitive.
- **G4** AXON T+30 outcome (~2026-06-08): first calibration data point; informs F6 timing.
- **G5** COHR decision after id=25 NO_REGIME_SHIFT: monitor for \$240-\$290 entry zone.

7. Near-Term Priority Sequence

Per the action sequence in `todo_master`:

1. **Now:** commit/push the session work + CHANGELOG entry (largely done).
 2. **Week 2:** D2 (EDGAR rewrite) and D1 (kill-rule routing, after V2 wiring) — can run in parallel after C1.
 3. **Parallel/ad-hoc:** E1-E3 quick data checks.
 4. **Week 3+:** D3 → D4 → D5 in order of conviction.
 5. **Watch 2026-06-08:** AXON T+30 → triggers the F6 position-sizing decision (G4).
 6. **Quarterly (~2026-08-26):** re-run the filter audit; fold in the G2 decision.
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8. Operating Principles (constraints on every change)

The codebase enforces a disciplined change philosophy worth preserving: one-step falsifiable changes (bundling causes contradictions); a complexity ratchet caution (a prior 17-fix episode *doubled* LITE's error — stay skeptical of magic numbers); the Hard-Gate / Contextual-Routing / Informational filter taxonomy for any new filter; a canonical 7-step workflow for non-trivial changes (light loop for surgical edits); milestone-level commit cadence with per-change CHANGELOG entries; and baseline-before-change discipline (the `v3.5-foundation` tag and basket snapshots exist precisely so downstream changes have unambiguous before/after evidence).

9. Bottom Line

The system is **healthy and operational**: scouts → analyst → engine → dashboard runs daily, the engine regression suite is green, and the two most ambitious recent additions (Socratic reasoning + macro/wave health) are live. The biggest structural bug (DCF undershoot on regime-shift names) is closed. The remaining work is mostly (a) **propagating** the DCF/archetype routing through the kill rule and the auto-promote heuristic, (b) **hardening data integrity** via an EDGAR-based guardrail, (c) **finishing the macro layer** with a cron orchestrator, and (d) **executing the frontend migration** to the production wireframe. Several reasoning refinements are deliberately parked until the first real outcome data (AXON T+30, ~2026-06-08) tells the system whether its recommendations are calibrated. The most time-sensitive item is not code at all — it is the LITE position decision at ~\$947 (G3).

Sources: `CLAUDE.md`, `CHANGELOG.md`, `data/todo_master_2026_05_26.md`, `config/watchlist.json`, `docs/design/MACRO_AND_WAVE_HEALTH_v1.md`, `docs/wireframes/migration-plan-v1.md`, `docs/reports/2026-05-*`, and live `scripts/` (engine tests run 2026-06-01: 37 passed).